

Kamil Hayyunaji Fabriano

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Fourth-year Geodesy and Geomatics Engineering student at Institut Teknologi Bandung with interests in GIS, hydrography, and spatial data analysis. Experienced in bathymetric mapping, LiDAR processing, and geospatial data analysis through internships, academic projects, and fieldwork, with proficiency in geospatial software, remote sensing, and surveying workflows.

EDUCATION

Institut Teknologi Bandung

2022 – Present

Bachelor of Geodesy and Geomatics Engineering (GPA: 3.36/4.00)

- Dean's List of Semester 3 – 2023/2024 and Semester 6 – 2024/2025
- Relevant Coursework: Hydrography, GIS, Photogrammetry, Remote Sensing, GNSS Survey, Spatial Database, Offshore Positioning, Nautical Sciences, Metocean Analysis, Hydrography Practices

WORK EXPERIENCE

Pertamina Hulu Rokan (PHR)

February 2026 – Present

Capstone Project – Development of a Datum Transformation Information System Using Framework Change Data and Plate Deformation Data in the Pertamina Hulu Rokan Environment

- Developed a web-based information system to automate datum transformation processes within Pertamina Hulu Rokan's operational area, integrating geodetic control network shift data and tectonic plate deformation models.
- Enabled accurate coordinate conversion between reference frames, supporting reliable spatial data management for oil and gas exploration and production activities.

Institut Teknologi Bandung

April 2026 – Present

Practicum Assistant (Artificial Intelligence and Data Literacy Course)

- Facilitated bi-weekly practicum sessions, guiding 63 students through modules covering descriptive statistics, scientific computing with NumPy and Matplotlib, spatial data visualization, and machine learning applications in earth sciences.

Badan Informasi Geospasial (BIG)

August 2025 – September 2025

Intern

- Integrated bathymetric data from primary hydroacoustic surveys (SBES, USV) with Satellite-Derived Bathymetry (SDB) from Sentinel-2B imagery to produce a 74,5 km² Digital Bathymetric Model (DBM) for the shallow waters of Laut Island, South Kalimantan.
- Evaluated the predictive accuracy of multiple machine learning models (Random Forest, K-Nearest Neighbors) for SDB against in-situ echosounder data, with the best-performing model (Random Forest) achieving RMSE of 0.38 m and analyzed the effects of atmospheric correction and algorithm parameters on model performance.

PT Tortuga Xcel Dynamics

January 2025 – February 2025

Intern

- Processed LiDAR point cloud data for solar farm site planning in Batang and Pemalang Regency, producing DTM, DSM, and contour maps.
- Conducted quality control on terrestrial survey data for a pipeline corridor survey (Pertamina Feed Pipeline RU III Plaju–Kertapati), verifying measurement accuracy and compliance with project specifications.

ORGANIZATIONAL EXPERIENCE

Ikatan Mahasiswa Geodesi Institut Teknologi Bandung (IMG-ITB)

Academic Tutor, Hydrography Expertise Group

June 2025 – December 2025

- Mentored junior students weekly in Hydrography Expertise Group courses, facilitating discussion sessions on course materials, problem sets, and practical surveying concepts.

Member, Student Representative Council (DPM) IMG-ITB

May 2025 – January 2026

- Developed and managed student aspiration platforms through bi-weekly surveys, interviews, and discussion forums to gather and analyze organizational evaluations.
- Delivered monthly processed aspiration reports to DPM IMG-ITB and executive boards of IMG-ITB for evaluation and improvement initiatives.

- Staff, Discussion, Research, and Literacy Department

July 2024 – April 2025

 - Contributed to spatial hazard, environmental assessment, and coastal policy studies related to marine spatial contestation and tidal flood risk through projects titled “*Marine Spatial Contestation in Coastal Areas: Policy Analysis and Social Reactions toward the Privatization of Sea Access*” and “*Spatial Assessment of Tidal Flood Risk in Jakarta Using Environmental Parameters*”
- Head, Entertainment Division Wisuda Oktober IMG-ITB 2024

June 2024 – January 2025

 - Directed a team of 11 staff in conceptualizing and delivering entertainment programs for IMG-ITB graduation celebrations.
 - Organized interactive activities to strengthen community bonding between students and graduates.
 - Oversaw the development of graduation appreciation media, including commemorative magazines and social media-based graduate spotlight campaigns.
- Staff, Professional Academy Department

October 2023 – April 2024

 - Facilitated practical training sessions for IMG-ITB members on surveying techniques using ETS, photogrammetry, and TLS, covering the full workflow from preparation and data acquisition to processing.

Apres! ITB

- Vice Head, Musical Arts Division

June 2024 – June 2025

 - Managed end-to-end audio production processes, including recording, mixing, and mastering songs created by Apres! ITB members.
 - Led and supervised 10 members in collaborative music album review and analysis projects within Apres! ITB.
 - Coordinated and guided 10 members in playlist curation projects tailored to thematic concepts selected by Apres! ITB members.

ADDITIONAL EXPERIENCES

Geopoint IMG-ITB

- Staff, Research and Development Division

August 2024 – May 2025

 - Acquired spatial data using GNSS and ETS for plant inventory mapping in Ikraland, Sumedang Regency.
- Staff, Data Processing Division

October 2023 – February 2024

 - Processed photogrammetric data using Agisoft Metashape to generate orthomosaic and 3D point cloud outputs for mapping purposes.

SKILLS

- GIS & Mapping:** ArcGIS Pro, QGIS, Global Mapper
- Photogrammetry & Remote Sensing:** Agisoft Metashape, ENVI, Spatix
- Surveying & Geodesy:** Bernese GNSS Software, Trimble Business Center, ETS, GNSS
- Hydrography & Oceanography:** SBES/MBES Processing, Satellite-Derived Bathymetry, Delft3D, MB System
- Programming & Data:** Python, MATLAB, PostgreSQL
- Other:** AutoCAD, Microsoft Office Suite

LICENCES AND CERTIFICATIONS

- Certified Drone Pilot**

Terra Drone Indonesia

(March 2026 – March 2028)
- Processing and Accuracy Assessment of SBES Data for 3D Modeling and Seafloor Topography**

Aksara Lab Indonesia

(February 2025)
- Hydrological Modeling Using SWAT+: Water Balance, Surface Runoff, and Sediment Erosion**

Geoaccess Indonesia

(February 2025)

LANGUAGES

- Bahasa Indonesia — Native
- English —EF SET 84/100, C2